

S'more Stoichiometry

Write a “balanced equation” for the production of s’mores. Use Gc for graham cracker, Mm for marshmallow, and Ch for chocolate.

Procedure:

1. Get one piece of each of the 3 items. Mass each of them individually and **record** the mass.
2. **Record** the mass (in grams) of the entire box/package of the three items.
3. **Calculate** how many pieces are in each package. (Don’t count them!)
4. **Record** how many of each piece are required for one s’more.
5. **Calculate** the maximum number of s’mores that can be made if you use the entire package of marshmallows, ignoring the other ingredients. Repeat for graham crackers and chocolate.

Data Table:

Food Item	Mass of Entire Package(s)	Mass of one piece	# of pieces in package(s)	# of pieces per s’more	Maximum number of s’mores based on this item
Marshmallow					
Chocolate					
Graham Cracker					

Questions:

1. Considering **all** of the ingredients, how many complete s’mores can you make?
2. Which item was limiting the number of s’mores you could make? This is called the limiting reagent.
3. Which items were in excess? These are called the excess reagents.
4. How much of the excess ingredients would be left over if you make the maximum number of s’mores?
5. How many chocolate pieces, graham crackers and marshmallows would you need to make 25 s’mores?
6. How many s’mores could you make if you started with 100.0 g of each reactant?

When this sheet is complete, you may perform a **synthesis reaction** to make a s’more.

